

EECS 489 - WN 25

Discussion 8

BGP, P3 Office Hours

Announcements

Finish Assignment 3 ASAP!

- Assignment 3 is **due Monday 3/31**.
- You have 3 late days in total for Assignment 2-4.
- Refer to Discussion 6 for tips on using UDP

Border Gateway Protocol

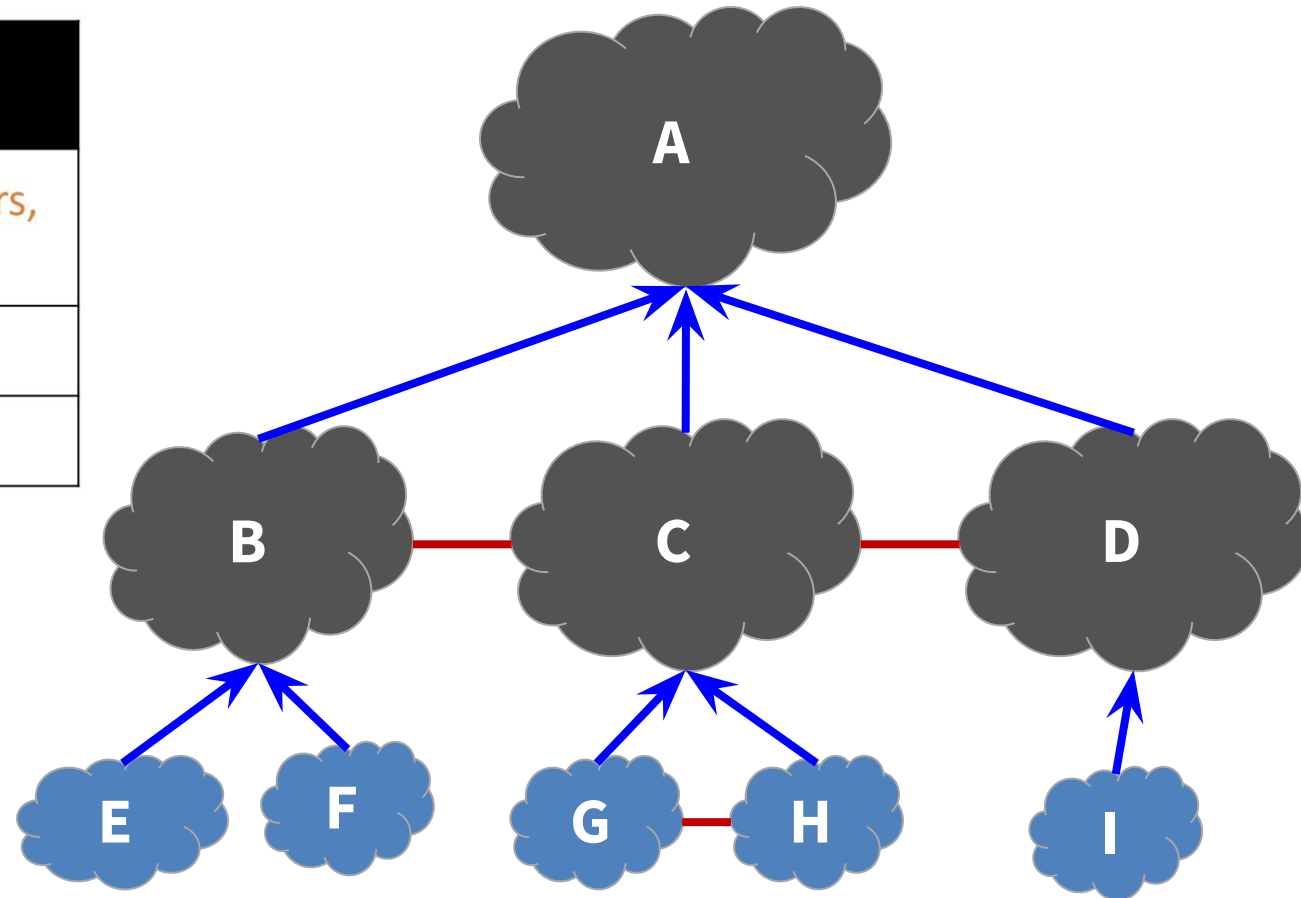
- Each **autonomous system** (AS) advertises its best routes to certain IP prefixes
- Other ASes select the best route that is advertised to them
- Two types of relationships between ASes:
 - **Provider-Customer:** Customer pays provider to carry customer traffic
 - **Peer-Peer:** No payment exchanged, route traffic through each other

Border Gateway Protocol

How to pick which routes to advertise? **Gao-Rexford Rules**

Destination prefix advertised by...	Export route to...
Customer	Everyone (providers, peers, other customers)
Peer	Customers
Provider	Customers

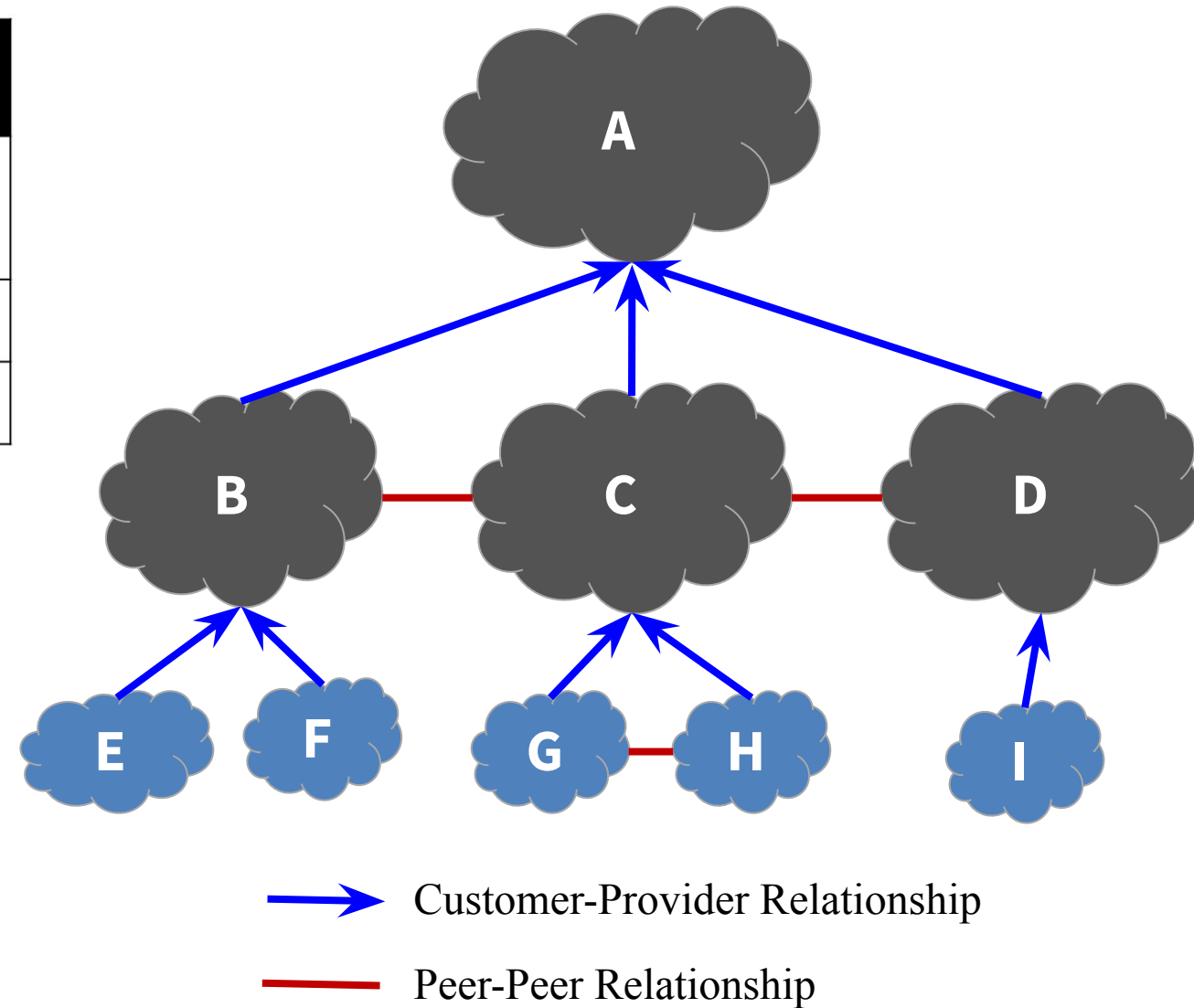
- Customer-Provider Relationship
- Peer-Peer Relationship



Exercise 1a: Gao-Rexford

Destination prefix advertised by...	Export route to...
Customer	Everyone (providers, peers, other customers)
Peer	Customers
Provider	Customers

What happens if E wants to talk to F?

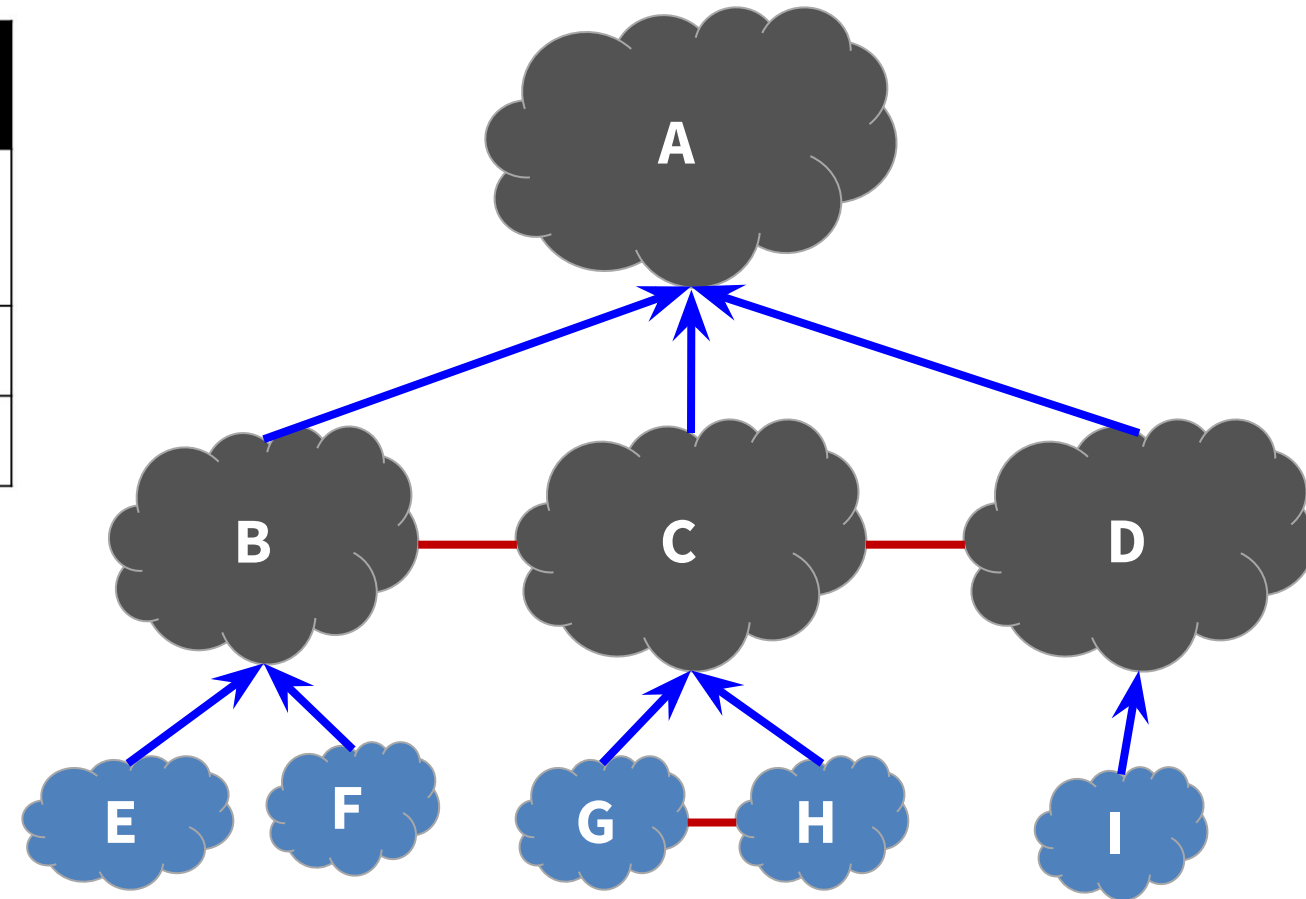


Exercise 1a: Gao-Rexford

Destination prefix advertised by...	Export route to...
Customer	Everyone (providers, peers, other customers)
Peer	Customers
Provider	Customers

What happens if E wants to talk to F?

$E \rightarrow B \rightarrow F$

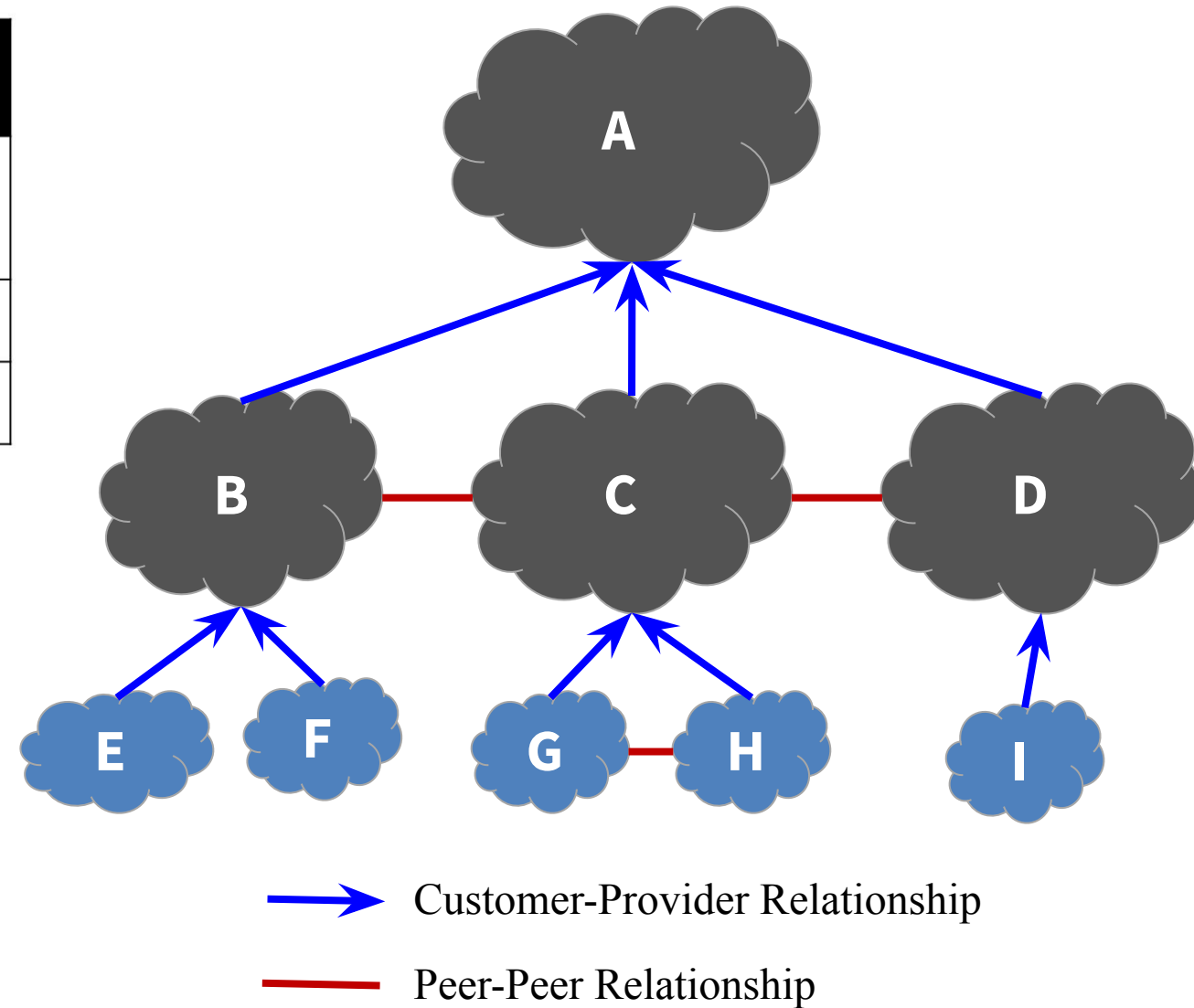


 Customer-Provider Relationship
 Peer-Peer Relationship

Exercise 1b: Gao-Rexford

Destination prefix advertised by...	Export route to...
Customer	Everyone (providers, peers, other customers)
Peer	Customers
Provider	Customers

What happens if F wants to talk to I?

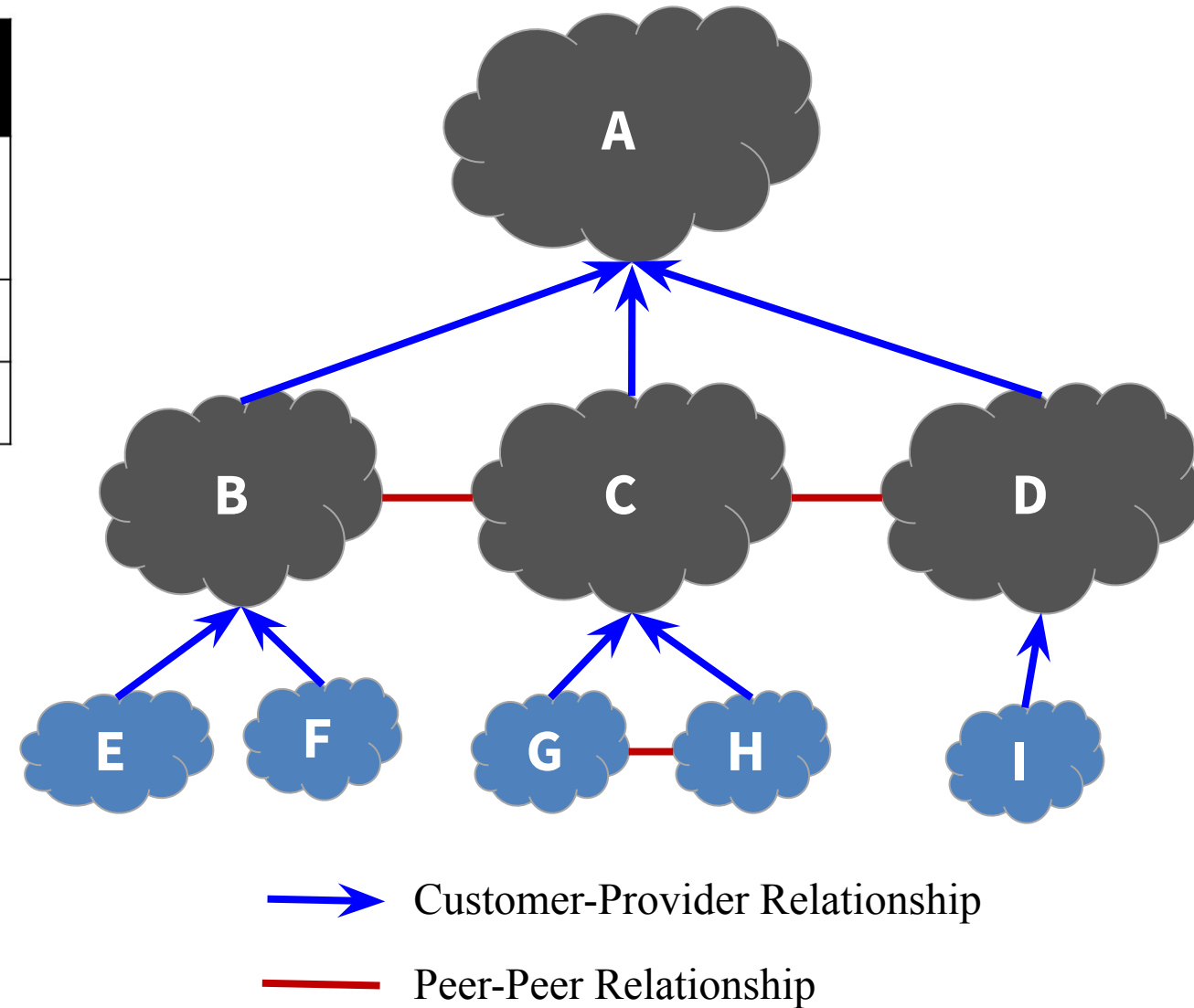


Exercise 1b: Gao-Rexford

Destination prefix advertised by...	Export route to...
Customer	Everyone (providers, peers, other customers)
Peer	Customers
Provider	Customers

What happens if F wants to talk to I?

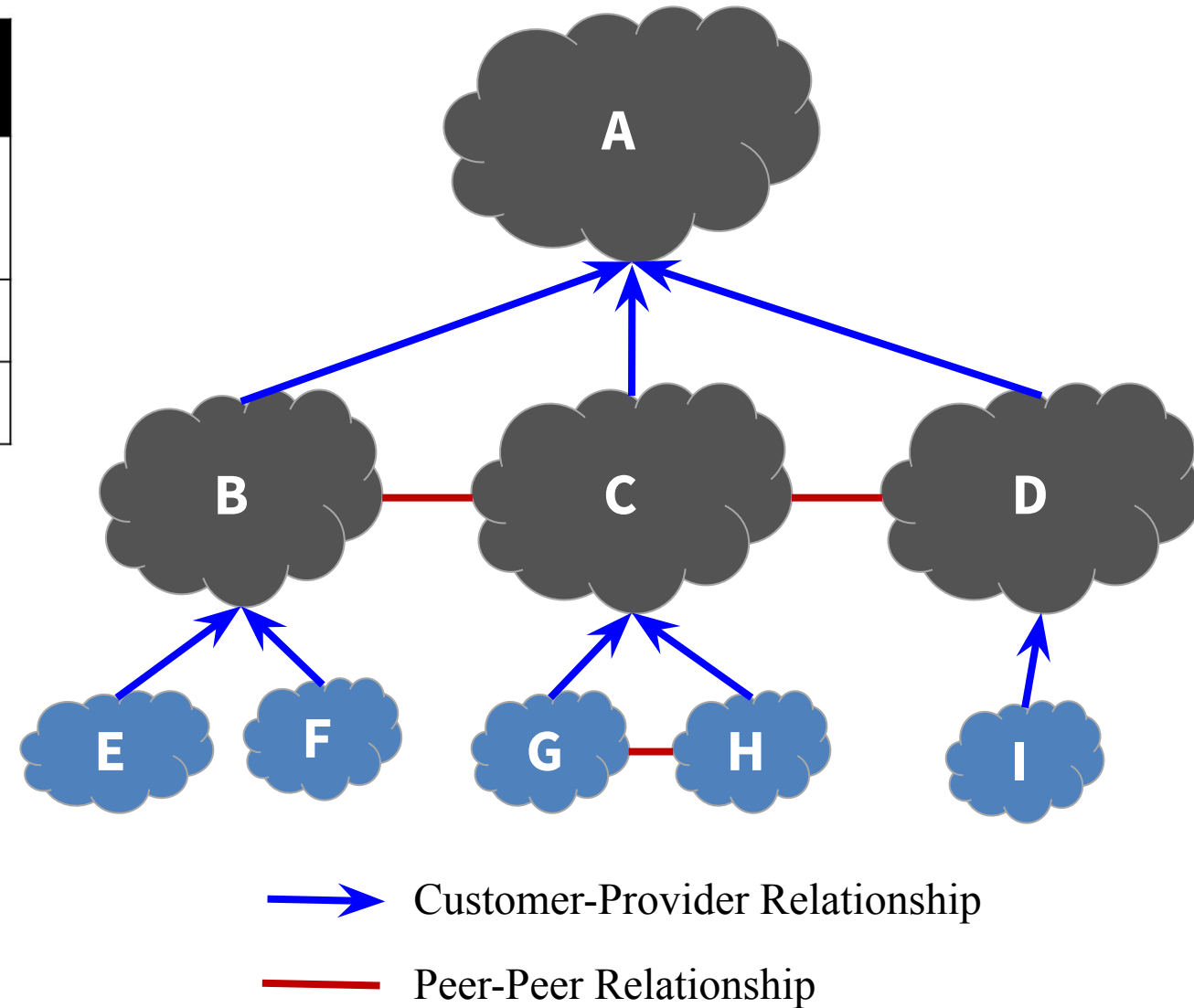
$F \rightarrow B \rightarrow A \rightarrow D \rightarrow I$



Exercise 1c: Gao-Rexford

Destination prefix advertised by...	Export route to...
Customer	Everyone (providers, peers, other customers)
Peer	Customers
Provider	Customers

What happens if E wants to talk to G?



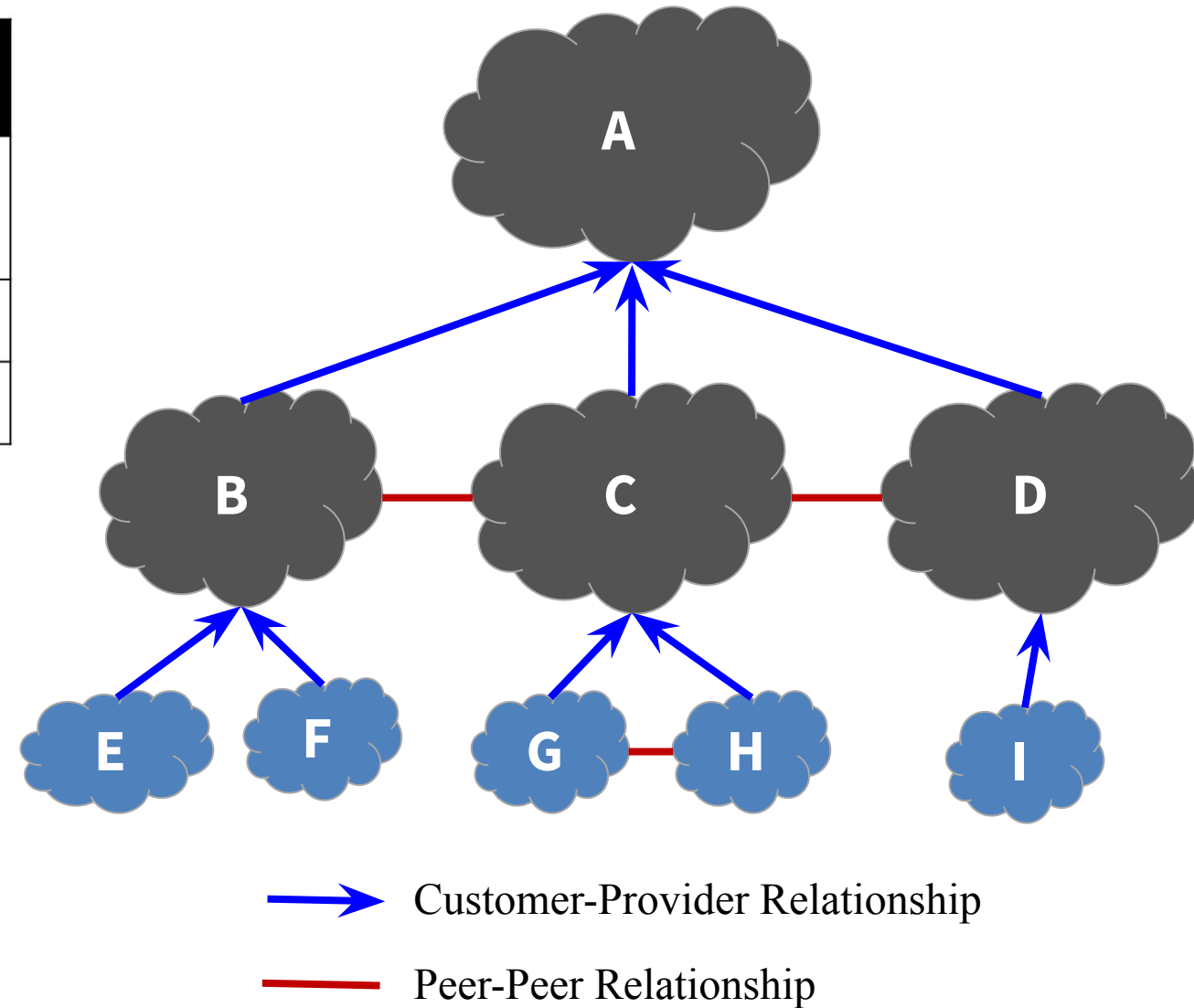
Exercise 1c: Gao-Rexford

Destination prefix advertised by...	Export route to...
Customer	Everyone (providers, peers, other customers)
Peer	Customers
Provider	Customers

What happens if E wants to talk to G?

$E \rightarrow B \rightarrow C \rightarrow G$.

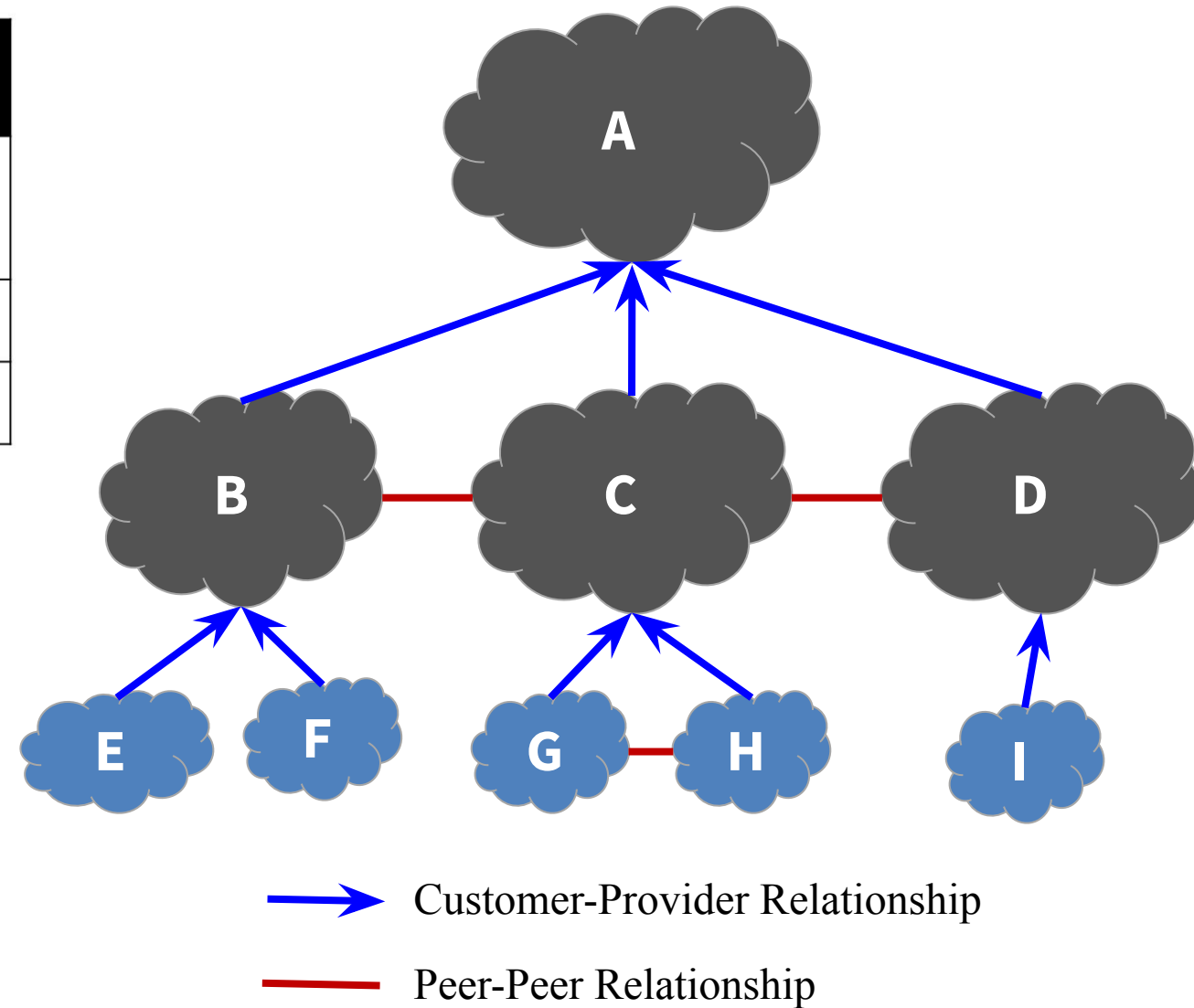
B would not go through A, since that would cause B to pay some cost to A.



Exercise 1d: Gao-Rexford

Destination prefix advertised by...	Export route to...
Customer	Everyone (providers, peers, other customers)
Peer	Customers
Provider	Customers

What happens if B wants to talk to D?



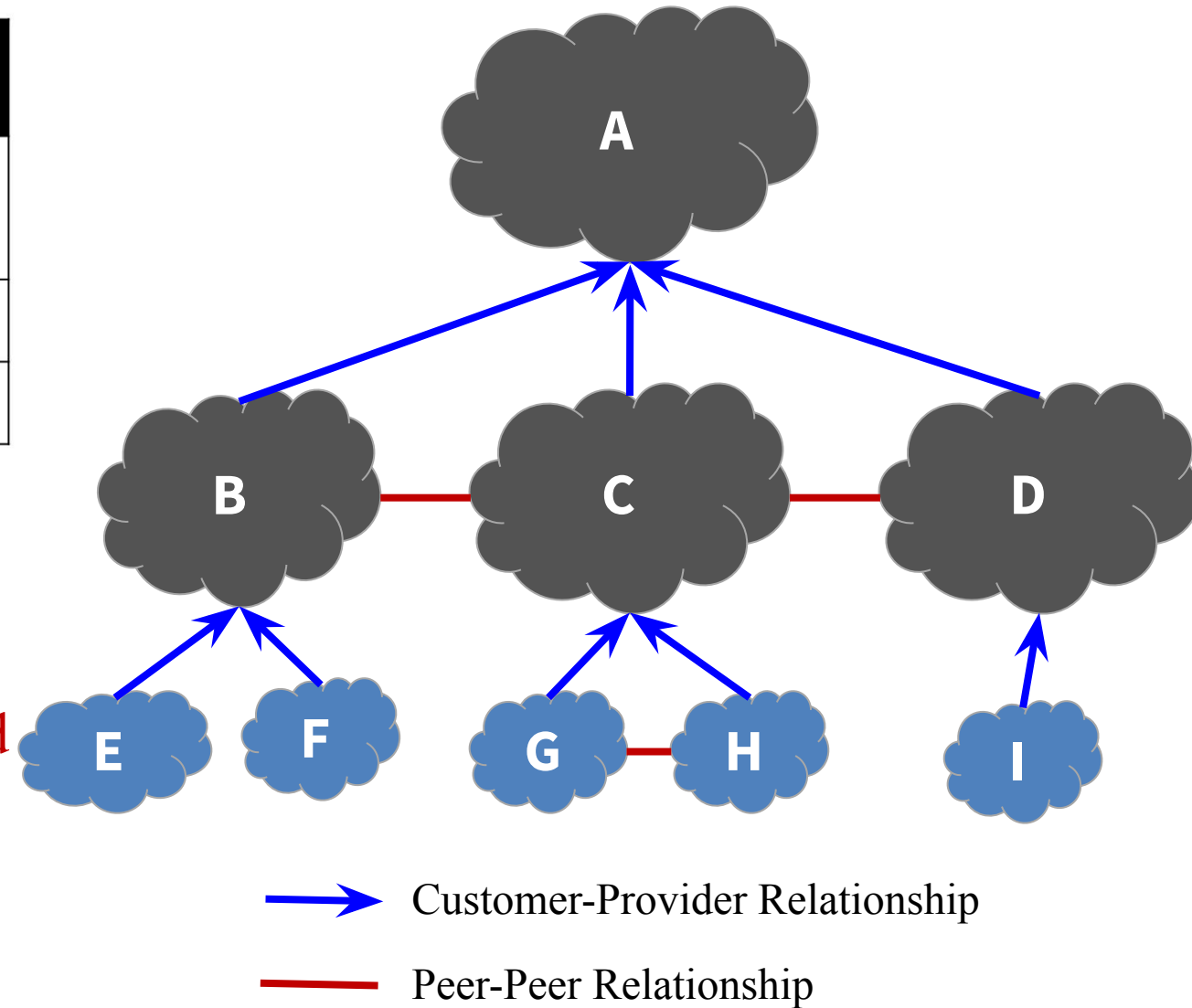
Exercise 1d: Gao-Rexford

Destination prefix advertised by...	Export route to...
Customer	Everyone (providers, peers, other customers)
Peer	Customers
Provider	Customers

What happens if B wants to talk to D?

$B \rightarrow A \rightarrow D$

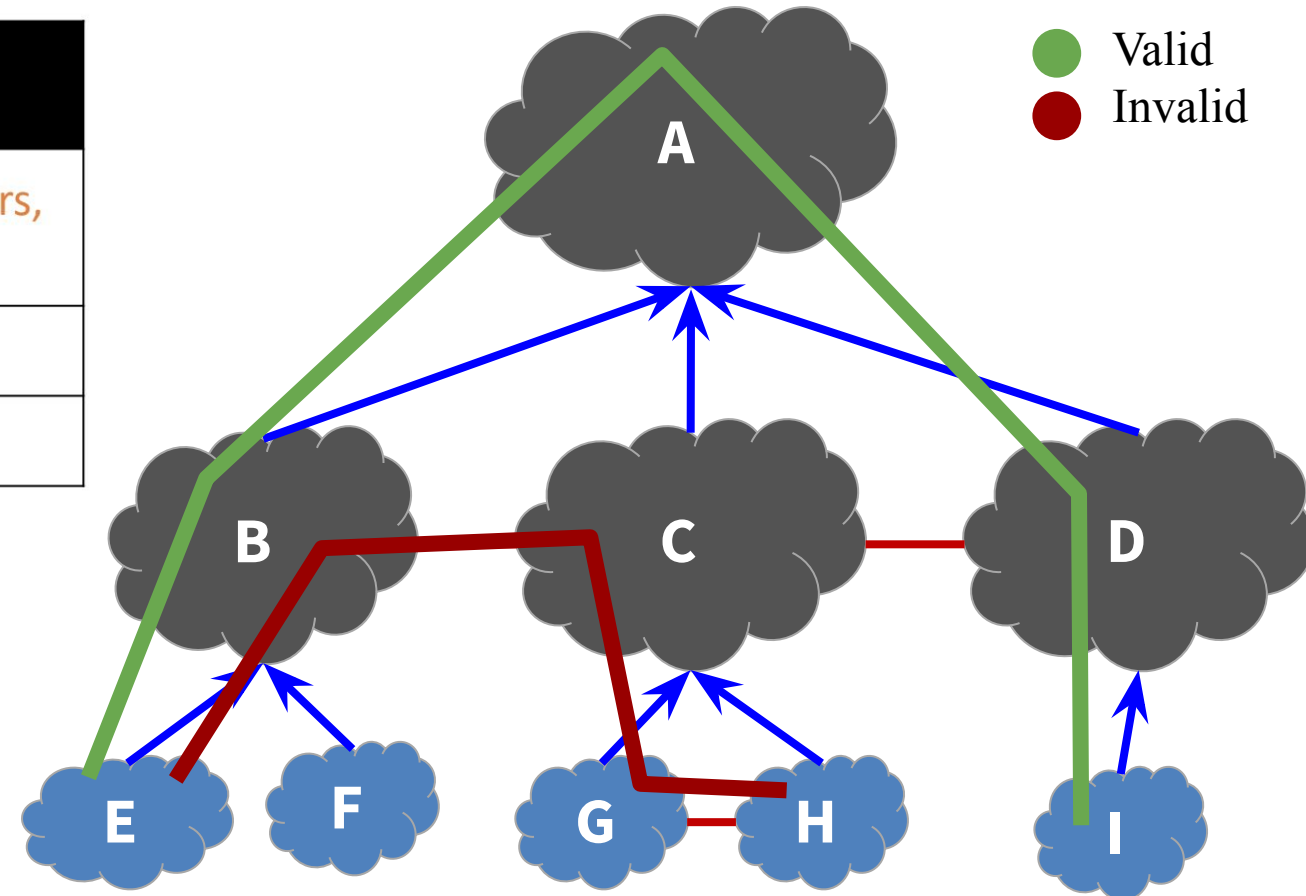
Note that $B \rightarrow C \rightarrow D$ does not work; C would not tell B that it has a path to D.



Border Gateway Protocol

Observe: Gao-Rexford leads to **valley-free** routes (never dip to customers prematurely).

Destination prefix advertised by...	Export route to...
Customer	Everyone (providers, peers, other customers)
Peer	Customers
Provider	Customers



Border Gateway Protocol

- eBGP is used by border routers to communicate across ASes
- iBGP is used to distribute eBGP information within an AS
- IGP is another name for intra-AS routing; it can be one of any number of protocols, including OSPF (link-state), RIP (distance-vector)

Border Gateway Protocol

Routes have **attributes** that describe them; these are the criteria for selecting a route.

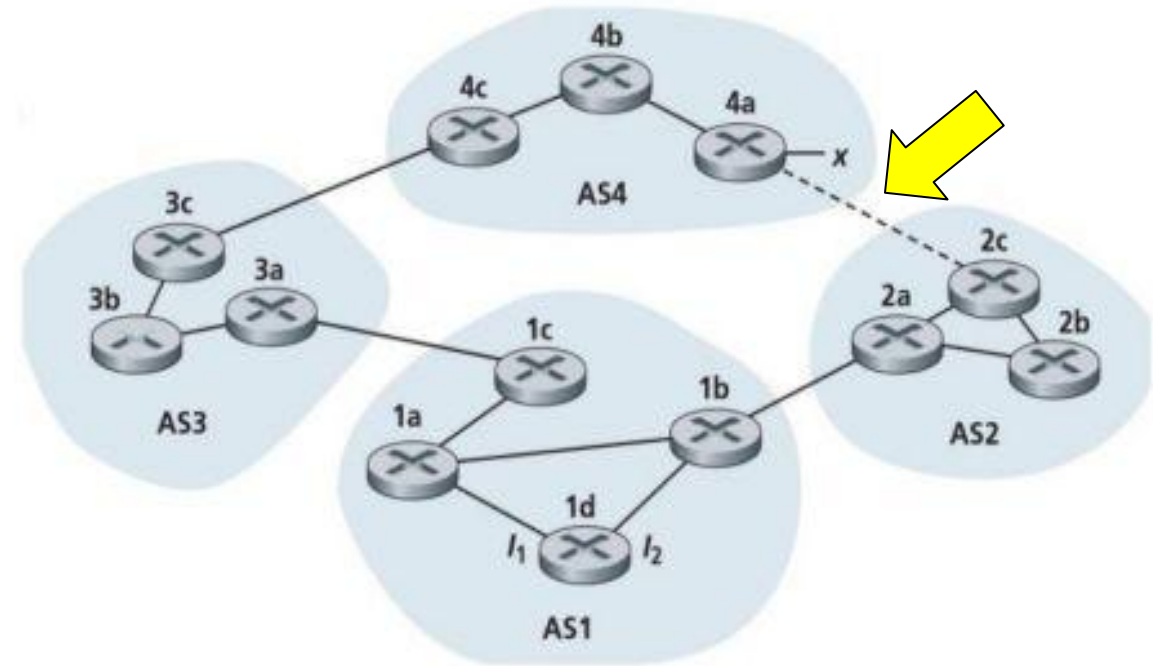
Priority	Rule	Description
1	LOCAL PREF	Indicates local preference (within an AS) of a path; essentially like an “override” for all the other attributes. Use Case: A network admin wants to use one route as the primary, and the other as a backup. Higher is better here.
2	ASPATH	Pick shortest ASPATH length (route through least number of ASes)
3	MED	When multiple links to next-hop AS exist, figure out which link is closer to the actual destination of the packet.
4	eBGP > iBGP	Prefer eBGP routes over iBGP as you want to get the packet out ASAP.
5	IGP cost	Which route gets the packet out fastest, based on IGP distances?
6	Router ID	Arbitrary tie-breaker.

Exercise 2a: BGP Attributes

Assumptions:

- No direct link between AS2, AS4.
- Every link within AS has the same cost.
- eBGP, iBGP being used.
- AS4 is provider to AS3
- AS3 is provider to AS1
- AS1 and AS2 are peers.

Which routing protocol (eBGP/iBGP/IGP) and router does Router 3c learn about prefix x from?



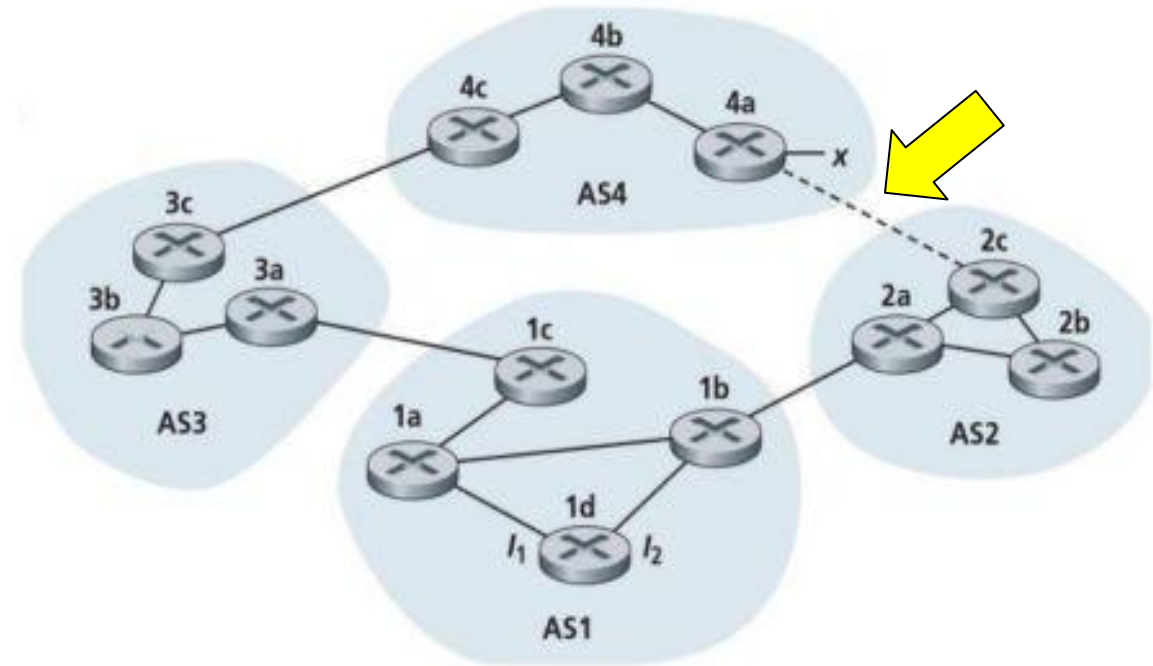
Exercise 2a: BGP Attributes

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Which routing protocol (eBGP/iBGP/IGP) and router does Router 3c learn about prefix x from?

eBGP from Router 4c

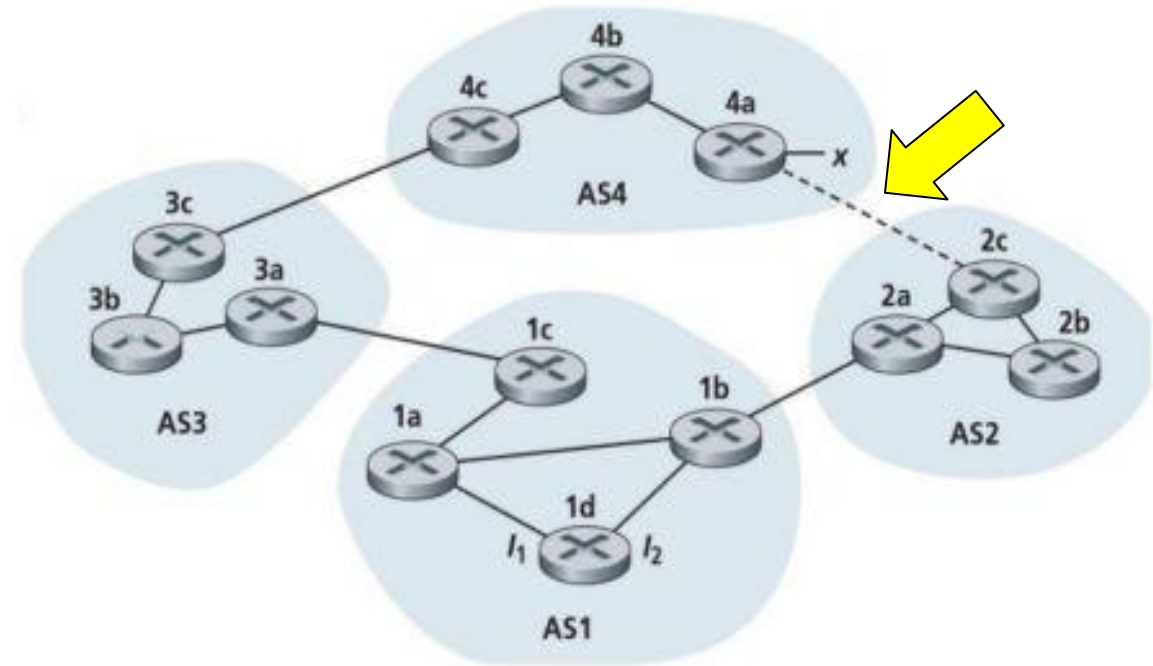


Exercise 2b: BGP Attributes

Assumptions:

- No direct link between AS2, AS4.
- Every link within AS has the same cost.
- eBGP, iBGP being used.
- AS4 is provider to AS3
- AS3 is provider to AS1
- AS1 and AS2 are peers.

Which routing protocol (eBGP/iBGP/IGP) and router does Router 4c learn about prefix x from?



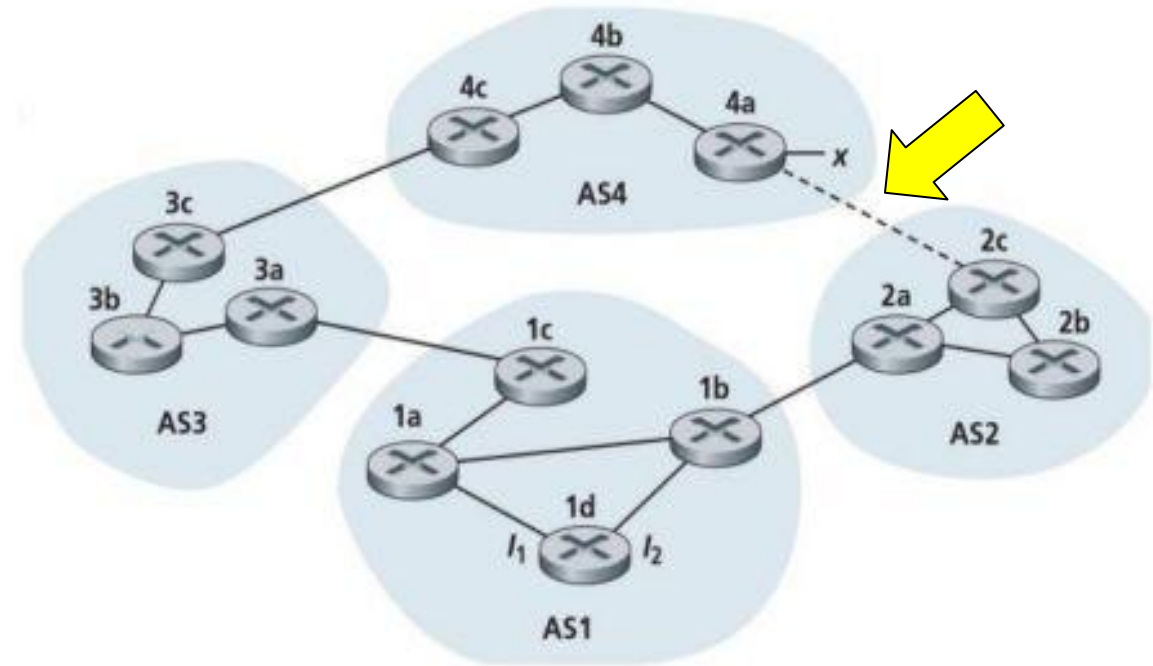
Exercise 2b: BGP Attributes

Assumptions:

- No direct link between AS2, AS4.
- Every link within AS has the same cost.
- eBGP, iBGP being used.
- AS4 is provider to AS3
- AS3 is provider to AS1
- AS1 and AS2 are peers.

Which routing protocol (eBGP/iBGP/IGP) and router does Router 4c learn about prefix x from?

IGP from Router 4b (note: not iBGP).

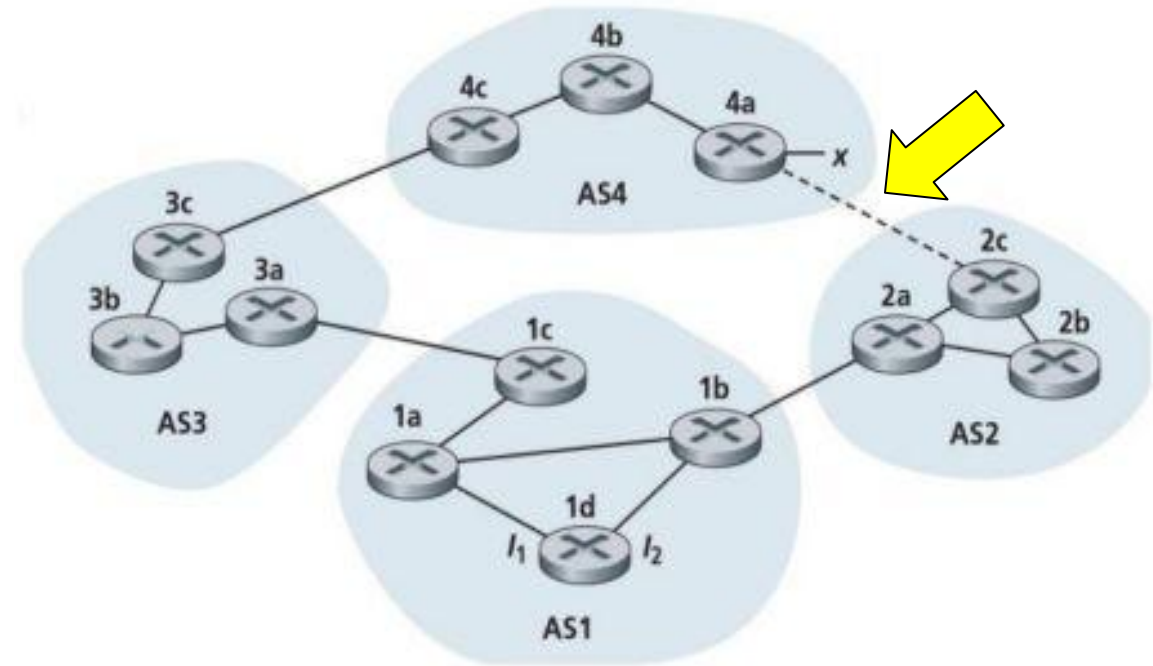


Exercise 2c: BGP Attributes

Assumptions:

- No direct link between AS2, AS4.
- Every link within AS has the same cost.
- eBGP, iBGP being used.
- AS4 is provider to AS3
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- AS1 and AS2 are peers.

Which routing protocol (eBGP/iBGP/IGP) and router does Router 1c learn about prefix x from?



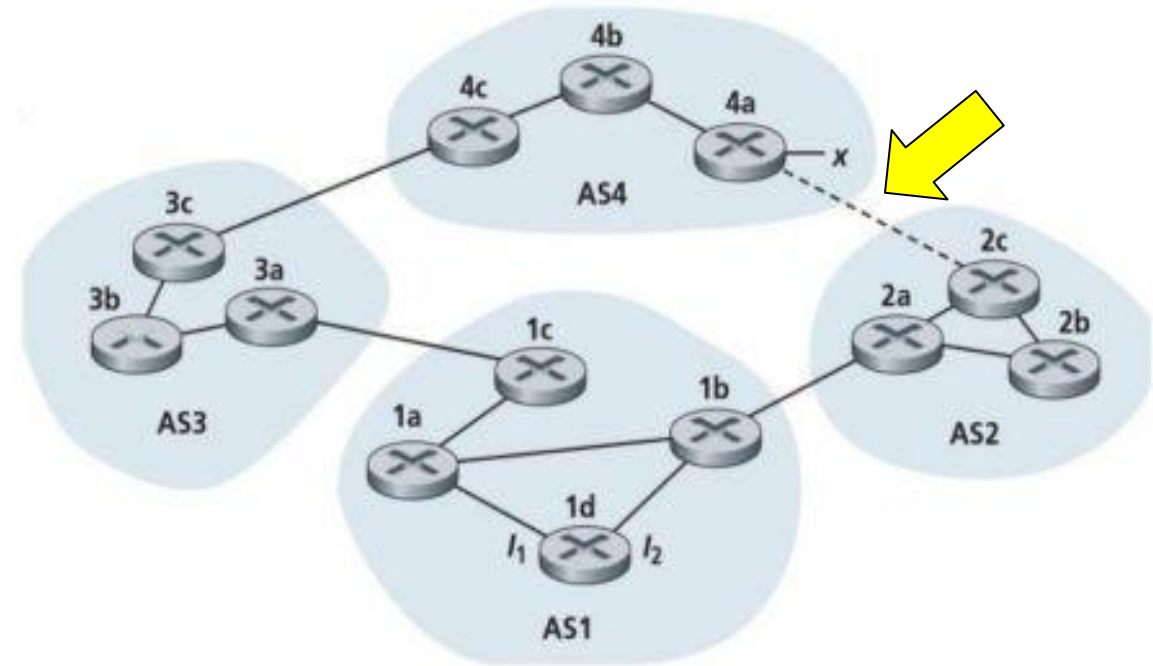
Exercise 2c: BGP Attributes

Assumptions:

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- eBGP, iBGP being used.
- AS4 is provider to AS3
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- AS1 and AS2 are peers.

Which routing protocol (eBGP/iBGP/IGP) and router does Router 1c learn about prefix x from?

eBGP from Router 3a.

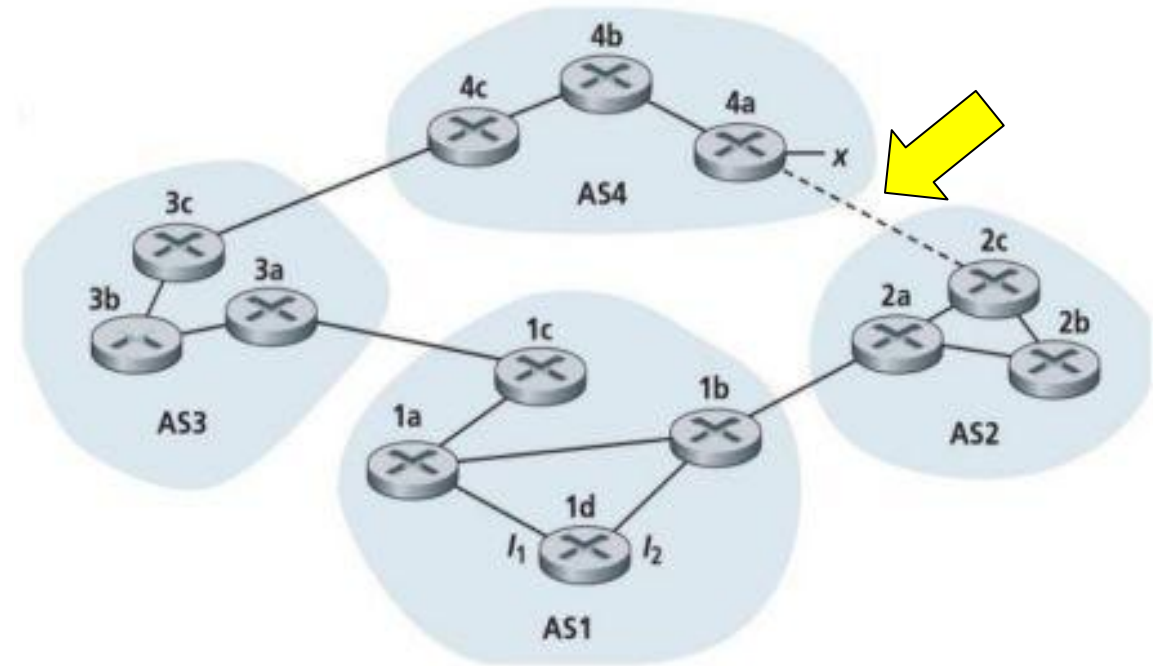


Exercise 2d: BGP Attributes

Assumptions:

- No direct link between AS2, AS4.
- Every link within AS has the same cost.
- eBGP, iBGP being used with standard export rules.
- AS4 is provider to AS3
- AS3 is provider to AS1
- AS1 and AS2 are peers.

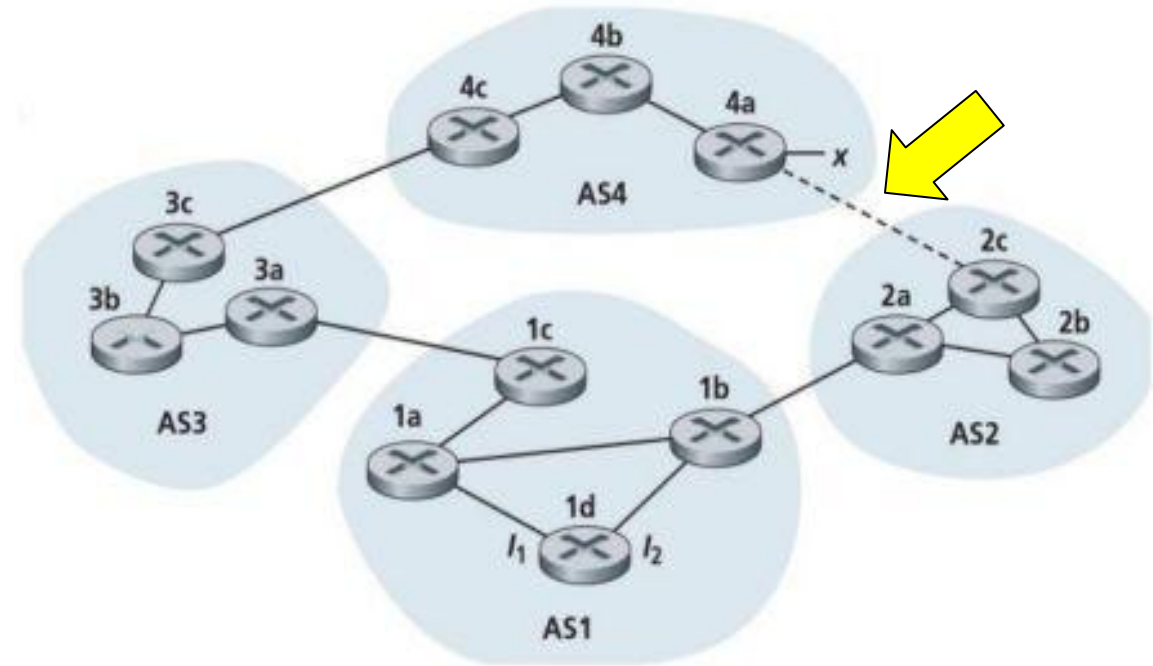
Can 2a reach prefix x under this configuration?



Exercise 2d: BGP Attributes

Assumptions:

- No direct link between AS2, AS4.
- Every link within AS has the same cost.
- eBGP, iBGP being used with standard export rules.
- AS4 is provider to AS3
- AS3 is provider to AS1
- AS1 and AS2 are peers.



Can 2a reach prefix x under this configuration?

No; as AS1 and AS2 are peers and AS3 is a provider to AS1, AS1 does not advertise the path to prefix x through AS3 to AS2. Thus, AS2 has no way of connecting to x.

Project 3 Questions

The remainder of the discussion will be office hours for Project 3.

Some general tips and things to consider when debugging:

Design your sender/receiver code to work for both base and OPT versions

- What differences are seen converting from base to OPT? What stays the same?
- Timing out?
 - Sender: how long should you wait before sending a new message, and what do you send?
 - Receiver: There should not be any waiting except calling `recvfrom()`. Make sure you handle every packet you receive properly! Any type of packet can be received at any time, how does this affect your state?
- Incorrect sequence numbers sent: go back to the protocol to consider when you send new packets and when you resend packets